

Arithmetic Sequences

Directions: Show all work. No calculators.

1. Determine if the sequence is arithmetic. If it is, find the common difference. 5, 11, 17, 23, ...

2. Determine if the sequence is arithmetic. If it is, find the common difference. 2, 5, 9, 14, ...

3. Determine if the sequence is arithmetic. If it is, find the common difference. 18, 13, 8, 3, ...

4. Determine if the sequence is arithmetic. If it is, find the common difference. 5, 10, 20, 40, ...

5. Determine if the sequence is arithmetic. If it is, find the common difference. 4, 9, 14, 19, ...

6. Determine if the sequence is arithmetic. If it is, find the common difference. $-3, 6, -12, 24, \dots$

7. Determine if the sequence is arithmetic. If it is, find the common difference. 31, 24, 17, 10, ...

8. Determine if the sequence is arithmetic. If it is, find the common difference. 2, 6, 12, 20, ...

9. Determine if the sequence is arithmetic. If it is, find the common difference. $-8, -8, -8, -8, \dots$

10. Find the common difference and write the next three terms: 21, 17, 13, 9, ...

11. Find the common difference and write the next three terms: $-12, -5, 2, 9, \dots$

12. Find the common difference and write the next three terms: $-25, -17, -9, -1, \dots$

13. Find a_{18} for the arithmetic sequence 7, 11, 15, 19,

14. Find a_{32} for the arithmetic sequence 50, 46, 42, 38,

15. Find a_{25} if $a_1 = 18$ and $d = -4$.

16. Which term of the arithmetic sequence 6, 11, 16, 21, ... is 101?

17. Which term of the arithmetic sequence 40, 37, 34, 31, ... is -11 ?

18. Write an explicit formula for the arithmetic sequence 28, 22, 16, 10,

19. An arithmetic sequence has $a_1 = -7$ and $d = 5$. Write the first five terms and an explicit formula.

20. A small theater has 24 seats in the first row. Each row behind it has 3 more seats than the row before. Write an explicit formula for the number of seats in row n , then find the number of seats in row 18.

21. A student saves \$45 during the first week and increases the amount saved by \$10 each week. Write an explicit formula for the amount saved during week n , then find the amount saved during week 12.